Lipoproteins

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Learning objectives

At the end of this lecture, you should be able to:

1. Classify lipoproteins according to their chemical compositions
2. Describe the various pathways of lipoprotein metabolism in detail
3. Know the role and importance of proteins involved in lipoprotein metabolism
4. Know the results that can arise from the disruption of lipoprotein metabolism

Outline

- Structure, composition and classification of lipoproteins

- Metabolism of lipoproteins
  - Chylomicron pathway
  - VLDL pathway
  - HDL and reverse cholesterol transport

References

1. Harvey R, Ferrier D. Lippincott's illustrated reviews: Biochemistry. 5th ed. Philadelphia: Lippincott Williams & Wilkins; 2011.

2. Murray RG, Granner DK, Mayes PA, Rodwell VW. Harper's illustrated biochemistry. 26th ed. New York: McGraw Hill Companies, Inc; 2003.

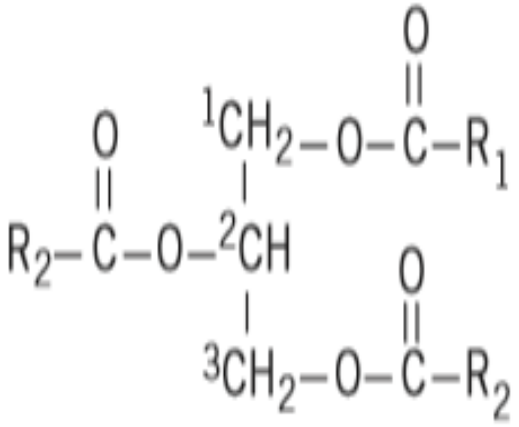
3. Smith CM, Marks AD, Lieberman MA. Marks' basic medical biochemistry: a clinical approach. 2nd ed. Lippincott Williams & Wilkins; 2005.

4. ดร.นพ. ภัทรบุตร มาศรัตน. 2557. เมตาบอลิซึมของไลโปโปรตีน. https://www.si.mahidol.ac.th/department/Biochemistry/home/MD/นศพ.57/เอกสารประกอบการสอนเรื่อง%20เมตาบอลิซึมของไลโปโปรตีน.pdf (สืบค้น 2 พฤษภาคม 2563)

List of important terms

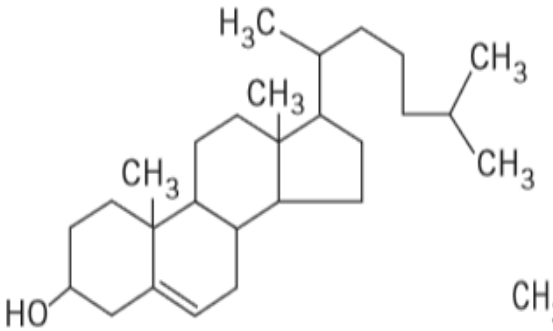
- Triglyceride (triacylglycerol)
- Cholesteryl ester (cholesterol ester)
- Lipoprotein
- Chylomicron
- Very low density lipoprotein (VLDL)
- Intermediate density lipoprotein (IDL)
- Low density lipoprotein (LDL)
- High density lipoprotein (HDL)
- Apolipoprotein AI (apo AI)
- Apolipoprotein B100 (apo B100)
- Apolipoprotein CII (apo CII)
- Apolipoprotein E (apo E)
- Lipoprotein lipase
- Hepatic lipase
- LDL receptor
- Lecithin cholesterol acyl transferase (LCAT)
- Cholesteryl ester transfer protein (CETP)

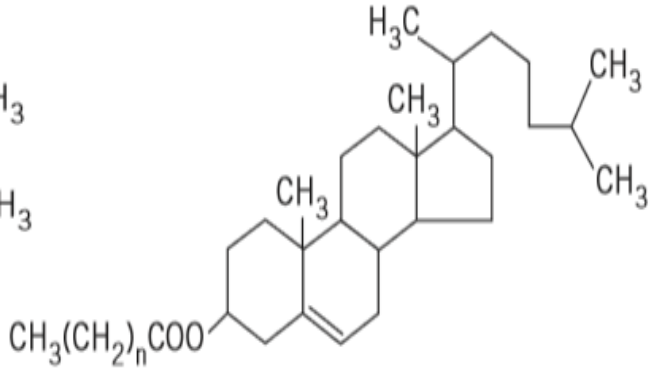
Major lipid classes found in lipoproteins



Triglyceride/Triacylglycerol (nonpolar lipid):

- ~ 90 % of dietary lipid
- Energy storage lipid in the body

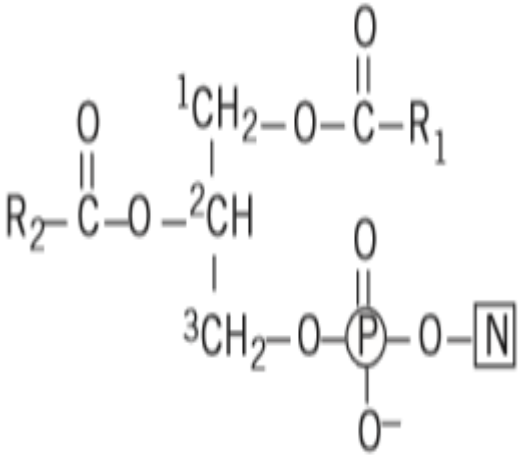




Cholesterol (amphipathic lipid):

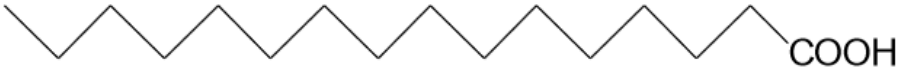
- Structural component of plasma membrane
- Precursor for synthesis of bile acid, hormones

Cholesteryl ester (nonpolar lipid)



Phospholipids (amphipathic lipid):

- Structural component of plasma membrane



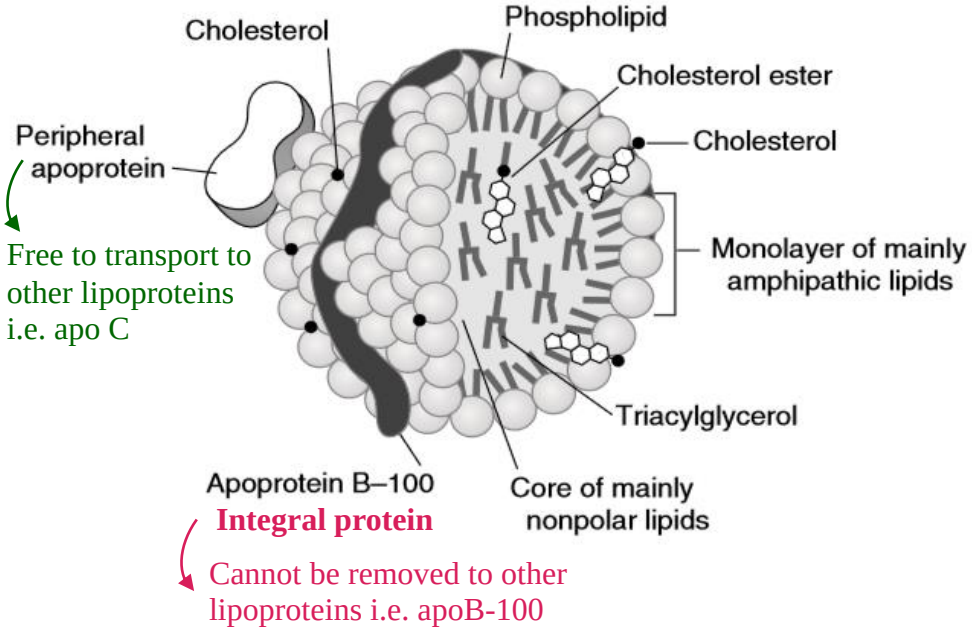
Fatty acids (amphipathic lipid):

- Component of triglyceride, phospholipids
- Precursor for eicosanoids

Lipoprotein structure

• Lipids absorbed from the diet and lipids synthesized by the liver and adipose tissue must be transported between the tissues for utilization and storage

• To transport lipids in the blood → nonpolar lipids associate with amphipathic lipids and proteins to make water-miscible lipoproteins



From Smith CM, Marks AD, Lieberman MA. Marks' basic medical biochemistry: a clinical approac. 2nd ed.. 2005.

Lipid:

- **Outer surface coat (**hydrophilic region)
  - Cholesterol
  - Phospholipid

- **Lipoprotein core** (hydrophobic region)
  - Triglyceride
  - Cholesteryl ester

Apolipoproteins:

- Divide into 5 classes (A to E) with each class has several subclasses i.e. apo AI, apo B48, apo CII
- Function of apoproteins:
  - Maintenance of lipoprotein structure
  - Enzyme cofactor or inhibitor i.e. apo CII (an activator of lipoprotein lipase)
  - Ligand for binding to lipoprotein receptors i.e. apo B-100 (target LDL receptor), apo E (target LDL receptor and LDL receptor related protein)

Composition of major human plasma lipoprotein classes

Lipoprotein	Source	Diameter (nm)	Density (g/mL)	Composition		Main Lipid Components	Apolipoproteins
				Protein (%)	Lipid (%)		
Chylomicrons	Intestine	90–1000	< 0.95	1–2	98–99	Triacylglycerol	A-I, A-II, A-IV,[1] B-48, C-I, C-II, C-III, E
Chylomicron remnants	Chylomicrons	45–150	< 1.006	6–8	92–94	Triacylglycerol, phospholipids, cholesterol	B-48, E
VLDL	Liver (intestine)	30–90	0.95–1.006	7–10	90–93	Triacylglycerol	B-100, C-I, C-II, C-III
IDL	VLDL	25–35	1.006–1.019	11	89	Triacylglycerol, cholesterol	B-100, E
LDL	VLDL	20–25	1.019–1.063	21	79	Cholesterol	B-100
HDL	Liver, intestine, VLDL, chylomicrons					Phospholipids, cholesterol	A-I, A-II, A-IV, C-I, C-II, C-III, D,[2] E
HDL_1		20–25	1.019–1.063	32	68		
HDL_2		10–20	1.063–1.125	33	67		
HDL_3		5–10	1.125–1.210	57	43		
Preβ-HDL[3]		< 5	> 1.210				A-I
Albumin/free fatty acids	Adipose tissue		> 1.281	99	1	Free fatty acids	

Abbreviations: HDL, high-density lipoproteins; IDL, intermediate-density lipoproteins; LDL, low-density lipoproteins; VLDL, very low density lipoproteins.
[1]Secreted with chylomicrons but transfers to HDL.
[2]Associated with HDL_2 and HDL_3 subfractions.
[3]Part of a minor fraction known as very high density lipoproteins (VHDL).

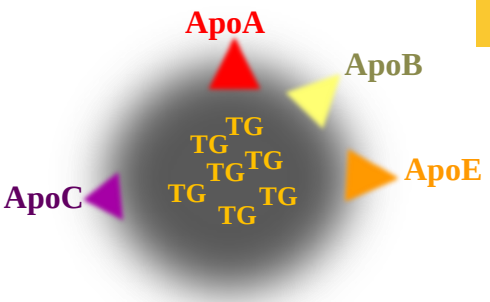
From Murray RG, Granner DK, Mayes PA, Rodwell VW. Harper's illustrated biochemistry. 26th ed. 2003.

Characteristics of major apoproteins

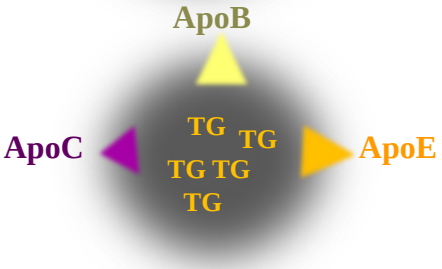
Apoprotein	Primary tissue source	Molecular mass (Daltons)	Lipoprotein distribution	Metabolic function
ApoA-1	Intestine, liver	28,016	HDL (chylomicrons)	Activates LCAT; structural component of HDL
ApoA-II	Liver	17,414	HDL (chylomicrons)	Unknown
ApoA-IV	Intestine	46,465	HDL (chylomicrons)	Unknown (may facilitate transport of other apoproteins between HDL and chylomicrons)
ApoB-48	Intestine	264,000	Chylomicrons	Assembly and secretion of chylomicrons from small bowel
ApoB-100	Liver	540,000	VLDL, IDL, LDL	VLDL assembly and secretion structured protein of VLDL, IDL, and LDL ligand for LDL receptor
ApoC-1	Liver	6,630	Chylomicrons, VLDL, IDL, HDL	Unknown; may inhibit hepatic uptake of chylomicron and VLDL remnants
ApoC-II	Liver	8,900	Chylomicrons, VLDL, IDL, HDL	Cofactor activator of lipoprotein lipase (LPL)
ApoC-III	Liver	8,800	Chylomicrons, VLDL, IDL, HDL	Inhibitor of LPL; may inhibit hepatic uptake of chylomicrons and VLDL remnants
ApoE	Liver	34,145	Chylomicron remnants, VLDL, IDL, HDL	Ligand for binding of several lipoproteins to the LDL receptor, to the LDL receptor-related protein (LRP) and possibly to a separate apo-E receptor.
Apo(a)	Liver		Lipoprotein "little" a (Lp(a))	Unknown

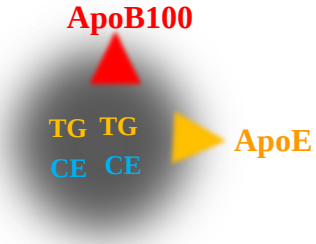
From Smith CM, Marks AD, Lieberman MA. Marks' basic medical biochemistry: a clinical approac. 2nd ed.. 2005.

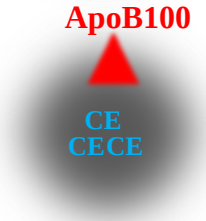
Composition of major human plasma lipoprotein



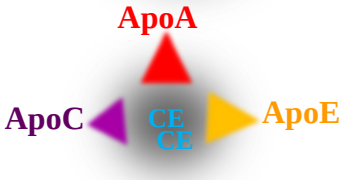
Larger size
Low density







Smaller size
High density



Prebeta HDL, HDL3, HDL2, HDL1

Chylomicron
98% lipid, 2% protein
Major lipid: triglyceride

Very low density lipoprotein (VLDL)
90% lipid, 10% protein
Major lipid: triglyceride

Transport triglyceride

Triglyceride from food to organs

Triglyceride from liver to organs

Intermediate density lipoprotein (IDL)
89% lipid, 11% protein
Major lipid: triglyceride, cholesterol

Low density lipoprotein (LDL)
79% lipid, 21% protein
Major lipid: cholesterol

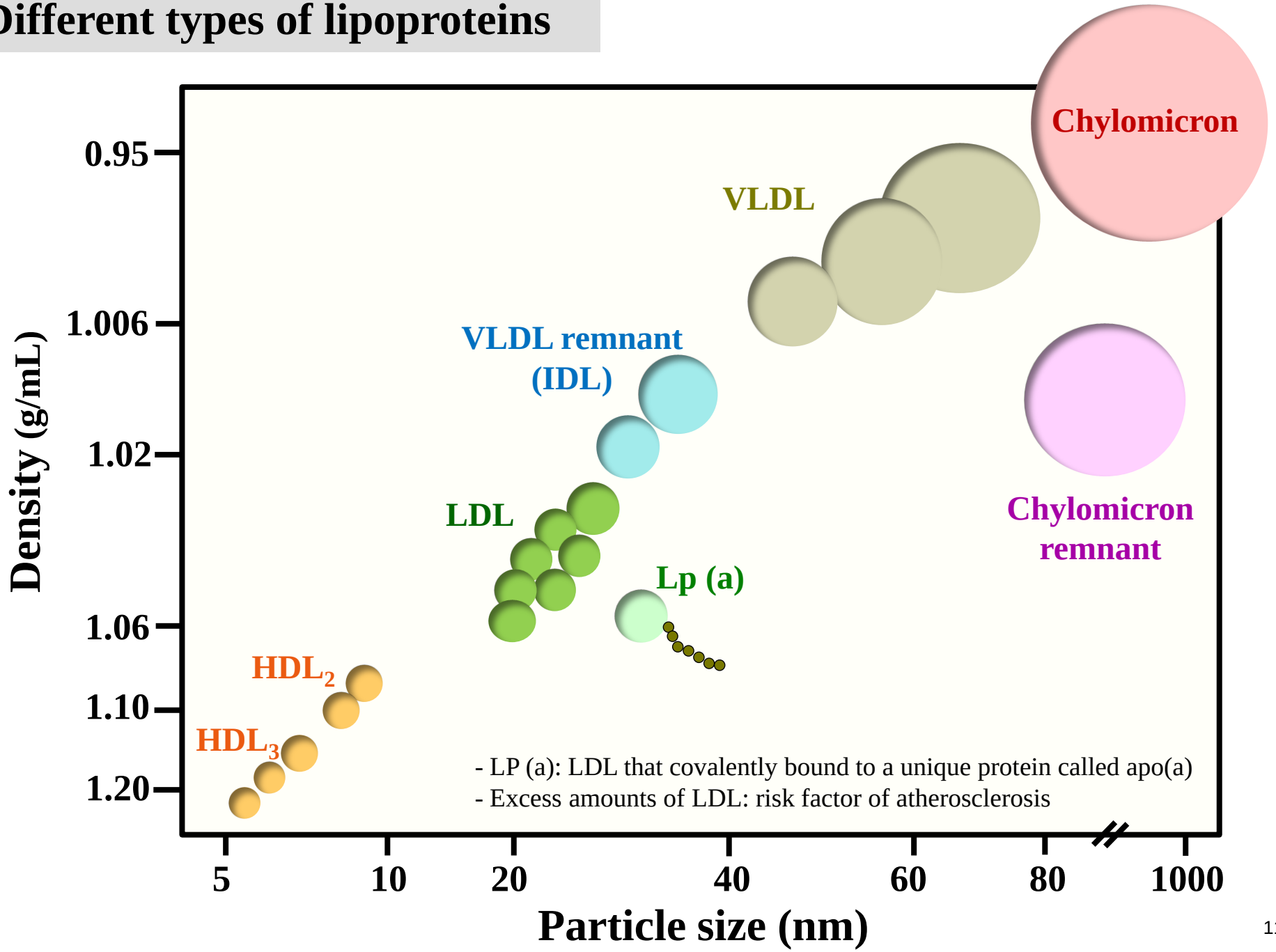
High density lipoprotein (HDL)
43-68% lipid, 32-57% protein
Major lipid: cholesterol

Transport cholesterol

Cholesterol from liver to peripheral tissues

Cholesterol from peripheral tissues to liver

Different types of lipoproteins



Lipoprotein metabolism

1. Exogenous pathway / chylomicron pathway:

- Carry all dietary lipids especially triglyceride (major lipid) from intestinal cells → tissues i.e. muscle, adipose tissue

2. Endogenous pathway / VLDL pathway:

- VLDL: carry endogenous triglyceride from the liver → peripheral tissues i.e. adipose tissue

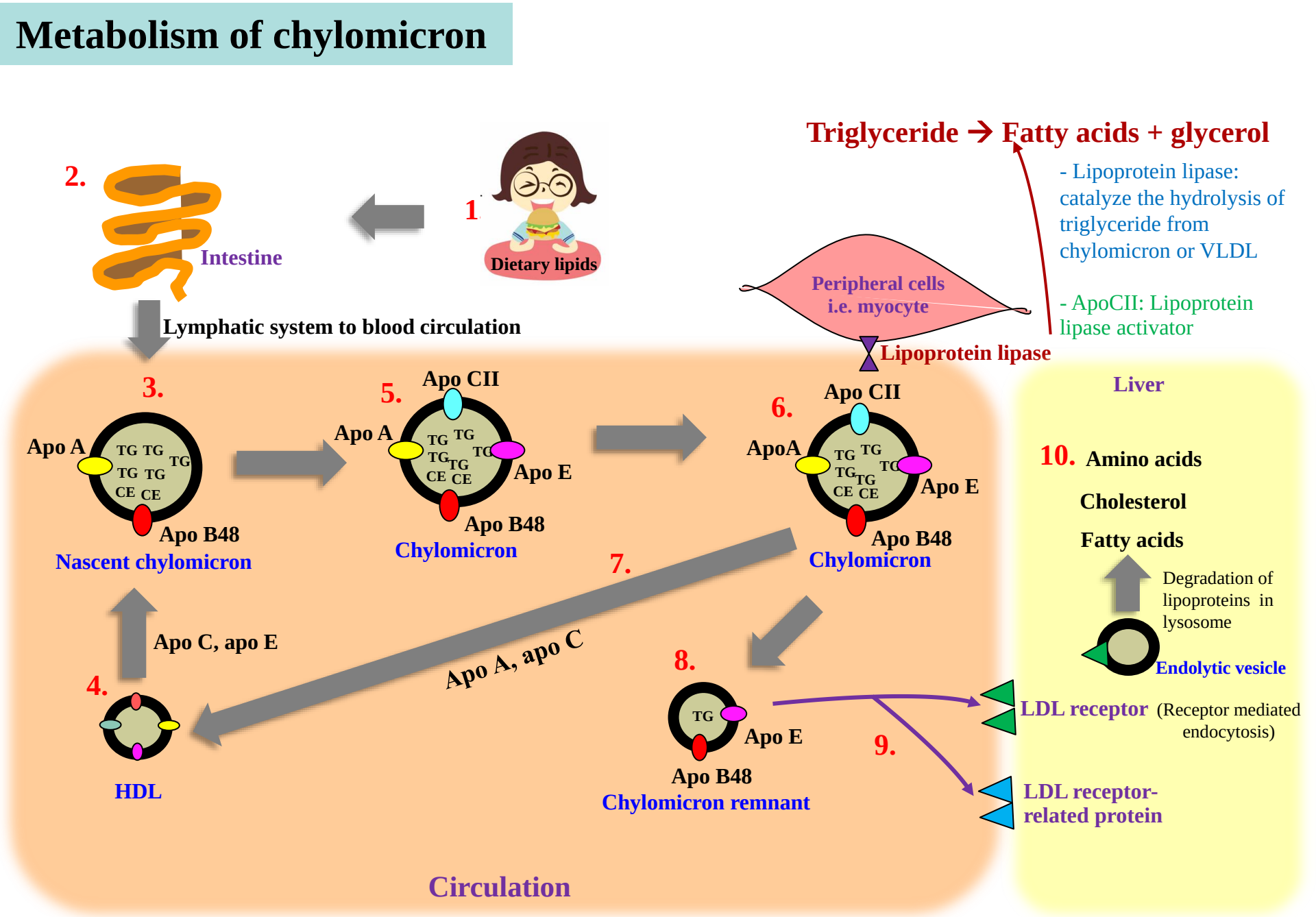
Factors that enhance the synthesis of triglyceride and the secretion of VLDL by the liver:
- The fed state rather than the starved state
- The feeding of diets high in carbohydrate → leading to high rates of lipogenesis
- High levels of circulating free fatty acids
- High concentrations of insulin and low concentrations of glucagon

- LDL: carry cholesterol → tissues

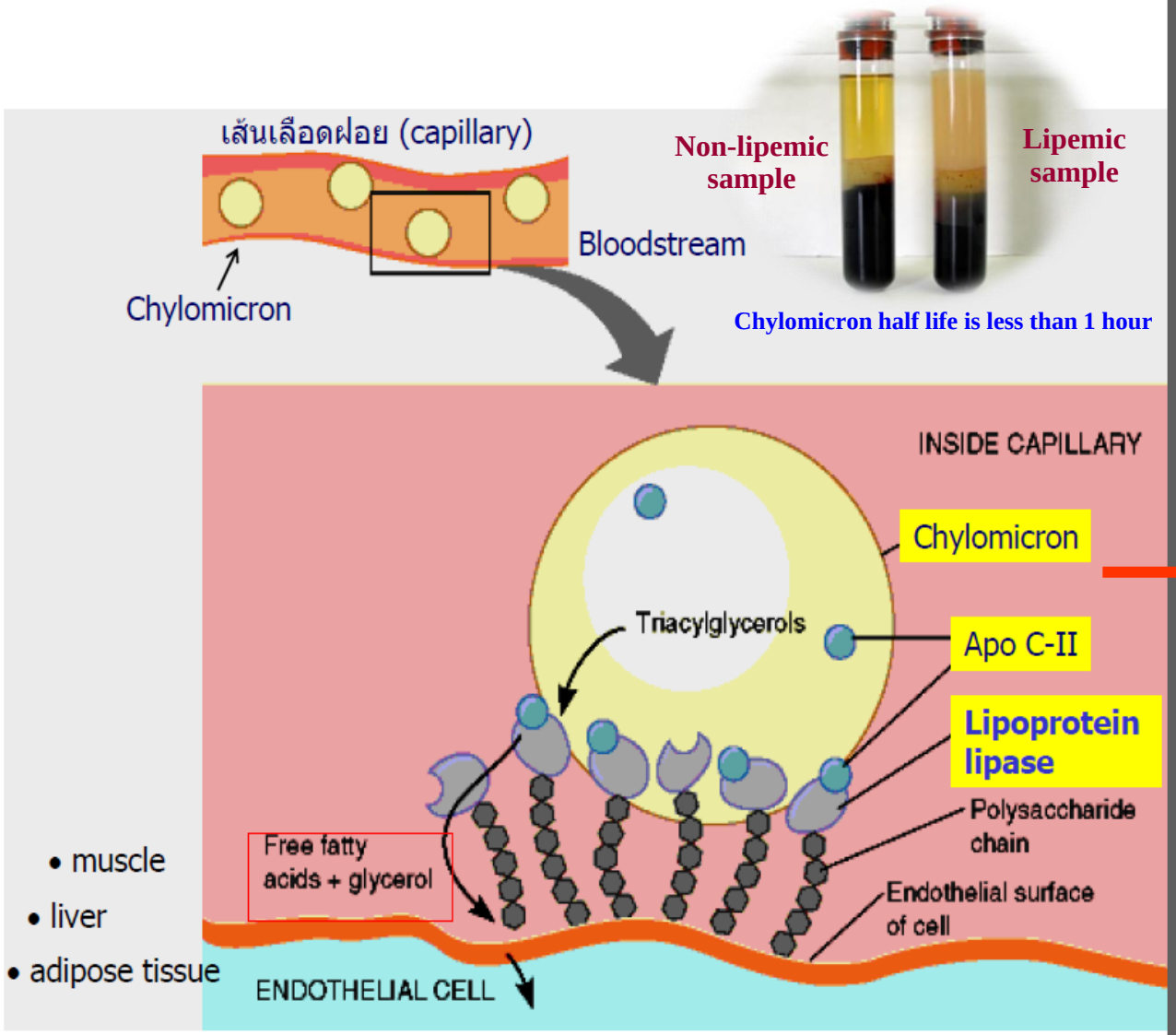
3. HDL metabolism:

- Picks up lipids from other lipoproteins
- Transfers apoprotein to other lipoproteins
- Picks up cholesterol from the body tissues and brings it back to the liver (reverse cholesterol transport) for excretion into the feces via the bile

Metabolism of chylomicron



Hydrolysis of triacylglycerol by lipoprotein lipase



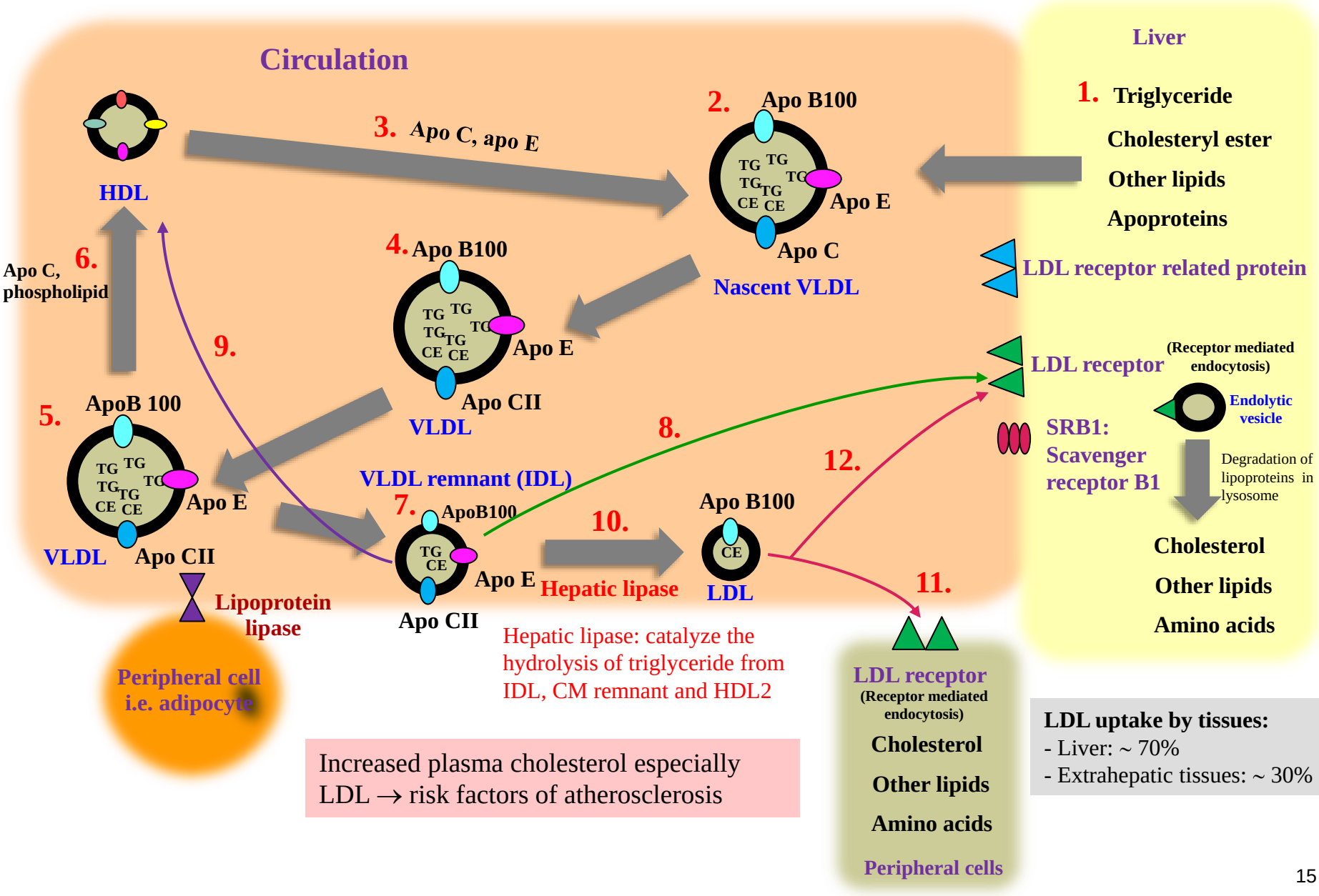
From Mathews CK, van Holde KE, Ahern KG. Biochemistry 3rd edition. 2000

Lipoprotein lipase or LPL

- Can be found in variety of tissues (especially in adipose tissue and striated muscle)
- Is secreted out of the cell and is located in the capillaries of the tissues
- Substrate: triglycerides and phospholipids of chylomicron and large VLDL

1. Chylomicron bind to LPL
2. Apo-CII activate the LPL
3. LPL hydrolyze the fatty acid ester bonds
4. The fatty acids are absorbed by the endothelial cells that line the capillary

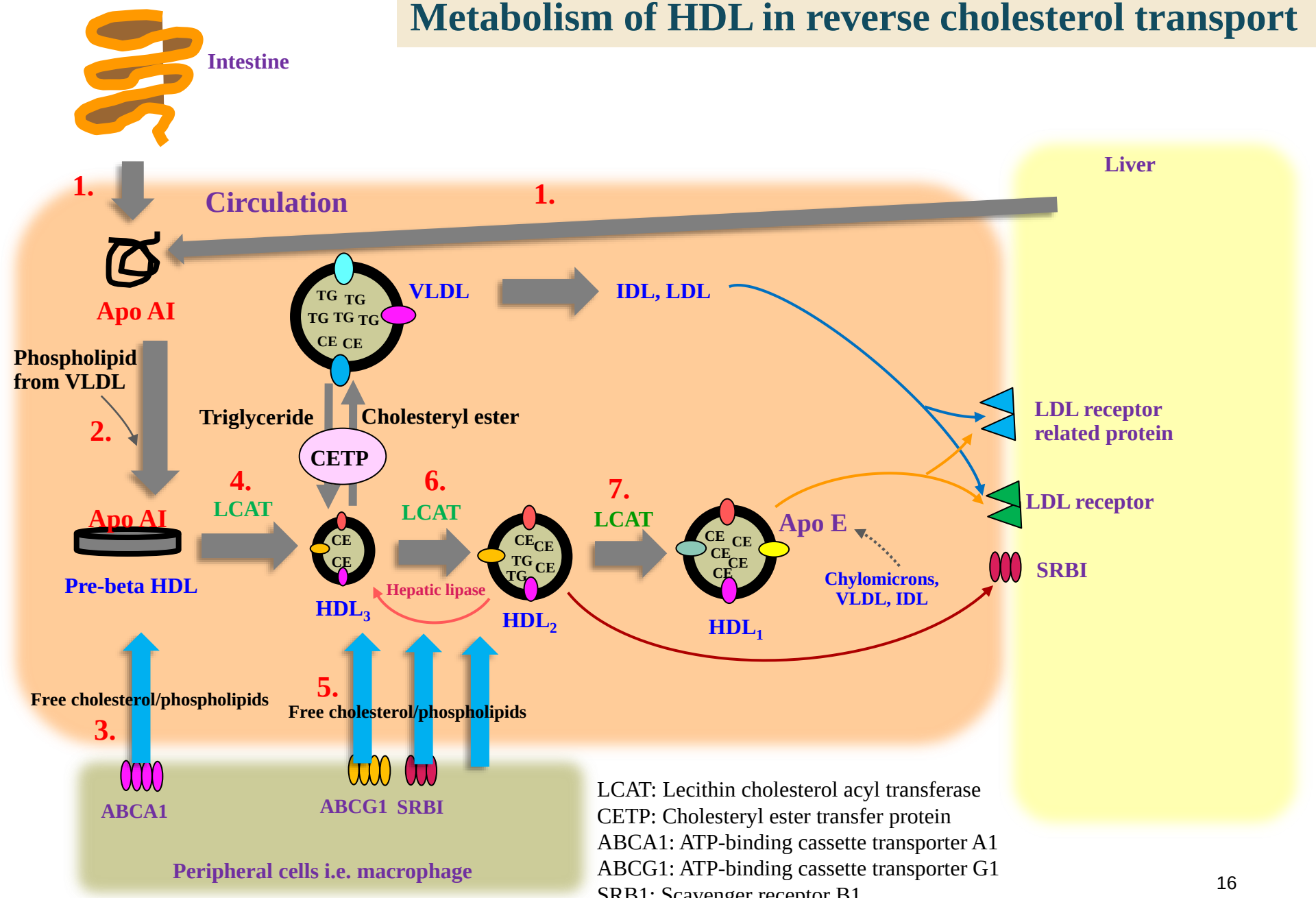
Metabolism of VLDL and LDL



Increased plasma cholesterol especially LDL → risk factors of atherosclerosis

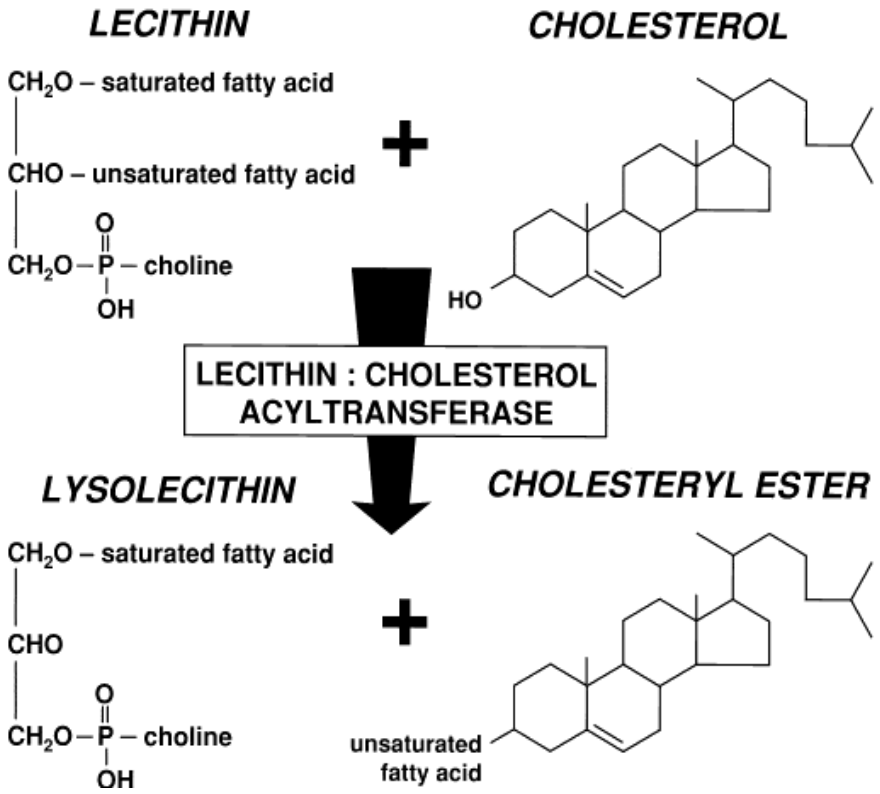
LDL uptake by tissues:
- Liver: ~ 70%
- Extrahepatic tissues: ~ 30%

Metabolism of HDL in reverse cholesterol transport



LCAT: Lecithin cholesterol acyl transferase
CETP: Cholesteryl ester transfer protein
ABCA1: ATP-binding cassette transporter A1
ABCG1: ATP-binding cassette transporter G1
SRB1: Scavenger receptor B1

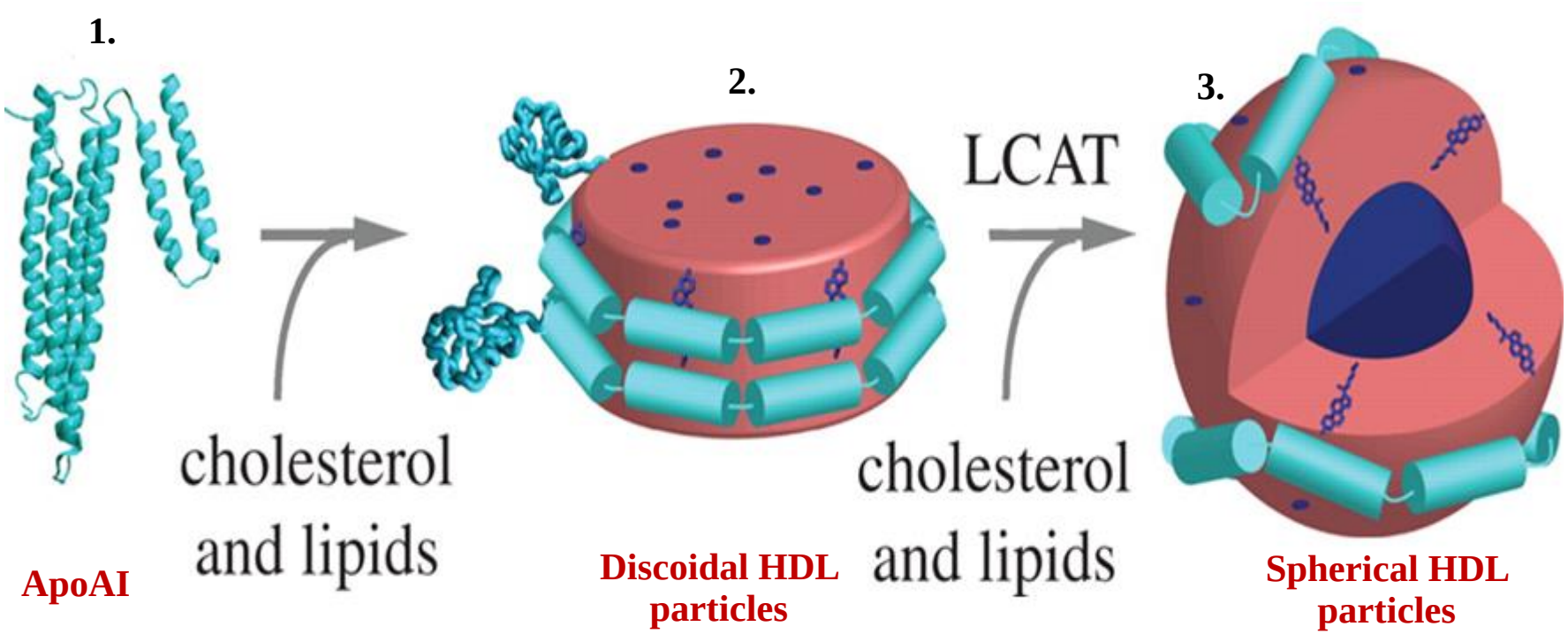
Lecithin cholesterol acyl transferase (LCAT):



From https://ommbid.mhmedical.com/content.aspx?sectionid=62637412&bookid=971

- Is synthesized in the liver and secreted into circulation
- Generates cholesterol ester by catalyzing the transfer of the acyl group of lecithin to the hydroxyl group of cholesterol
- HDL is the preferred lipoprotein substrate than LDL due to the presence of apoA-I (cofactor of LCAT)
- In humans, LCAT inactivating mutations lead to low plasma HDL, sharply increased LDL triglycerides, and accumulation of discoidal HDL in plasma

Two characteristic examples of lipoprotein structures (discoidal HDL particles and mature HDL)



From Shih, AY, Sligar, SG and Schulten, K. 2009

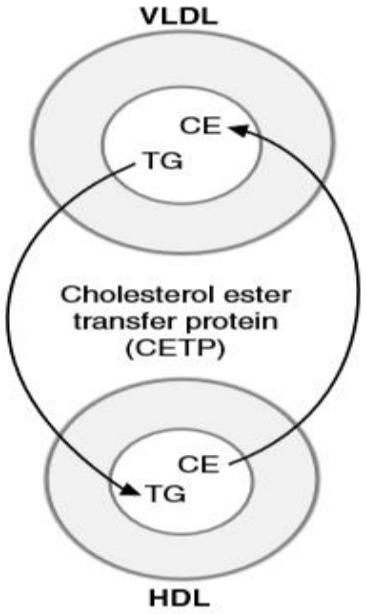
1. Lipid-free apo A-I is synthesized in the liver and intestines and secreted into the blood

2. ApoAI absorbs cholesterol and lipids forming discoidal HDL particles

3. Esterification of cholesterol into cholesterol ester by lecithin cholesterol acyltransferase (LCAT) resulting in the formation of spherical HDL particles.

Cholesteryl ester transfer protein (CETP):

From Smith CM, Marks AD, Lieberman MA. Marks' basic medical biochemistry: a clinical approach. 2nd ed.. 2005.



- Transport of cholesteryl esters and triglycerides between the lipoproteins (collects triglycerides from VLDL or LDL and exchanges them for cholesteryl esters from HDL), and vice versa)

- Play a major role in the catabolism of HDL (influences the concentration, apolipoprotein content, and size of HDL particles in plasma)

Hepatic lipase:

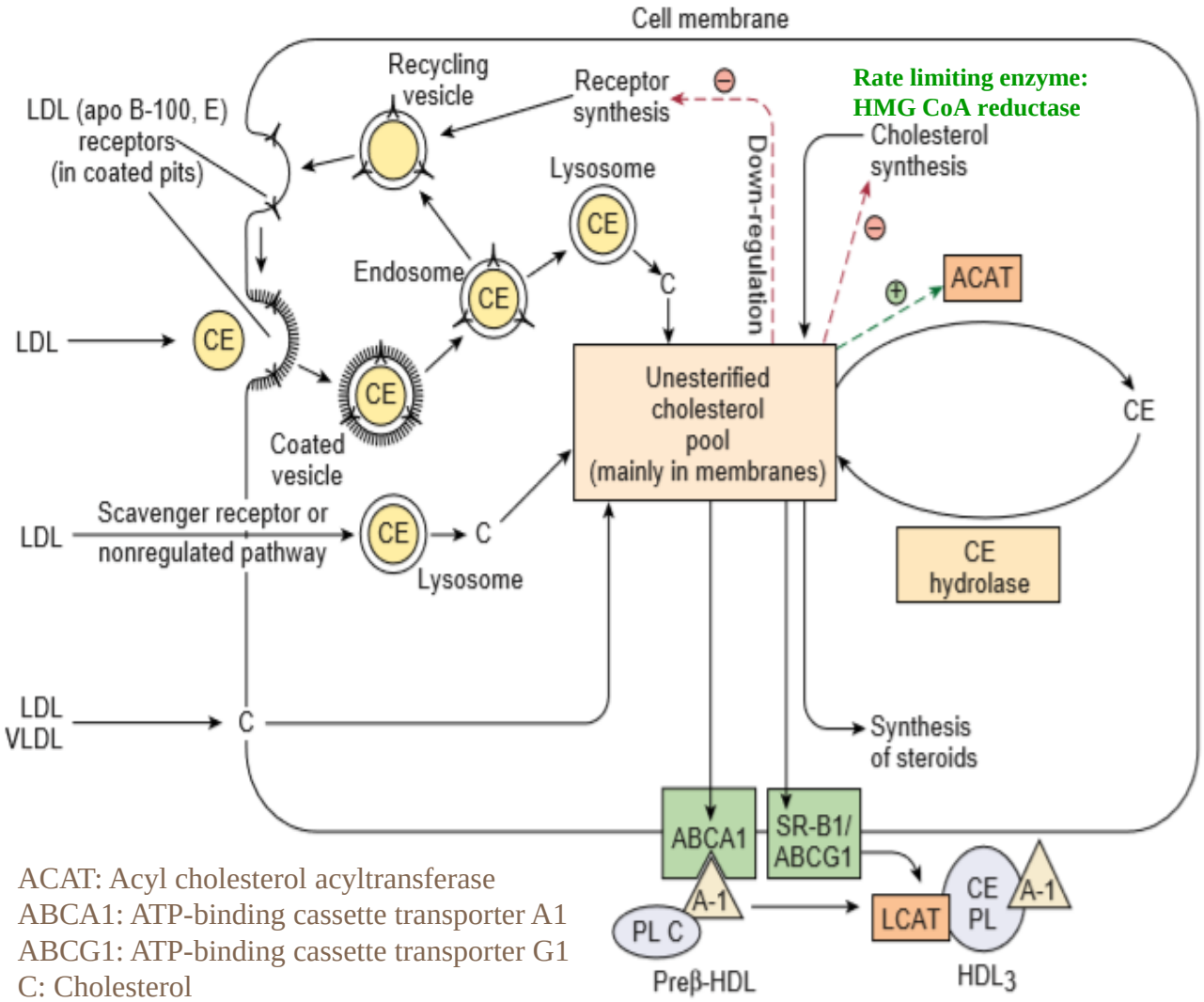
- Is synthesized in hepatocyte and is transported to the cell surface and to cells in the adrenal and gonad
- Involve in remnant and HDL metabolism

1. Act as a ligand that facilitates the uptake of lipoproteins by cell surface receptors (LDL receptor (LDLR), LDL receptor related protein (LRP) and scavenger receptor B1 (SR-B1)

2. Hydrolyses triglycerides and phospholipids in chylomicron remnants, IDL, and HDL (converting the HDL_2 to $HDL_{3)}$

Cholesterol balance at the cellular level

From Murray RG, Granner DK, Mayes PA, Rodwell VW. Harper's illustrated biochemistry. 26th ed. 2003.



ACAT: Acyl cholesterol acyltransferase
ABCA1: ATP-binding cassette transporter A1
ABCG1: ATP-binding cassette transporter G1
C: Cholesterol
CE: Cholesteryl ester
PL: phospholipid
SR-B1: Scavenger receptor B1
A-1: Apo A-I

➢ **Cell cholesterol increase is due to:**

- Uptake of cholesterol containing lipoproteins by LDL receptor, SRB1
- Endogenous synthesis of cholesterol (rate limiting enzyme is HMG CoA reductase)
- Hydrolysis of cholesteryl esters by cholesteryl ester hydrolase (CE hydrolase)

➢ **Cell cholesterol decrease is due to:**

- Efflux of cholesterol from the membrane to HDL via the ABCA1, ABCG1 or SR-B1
- Esterification of cholesterol to cholesteryl ester by Acyl CoA:cholesterol acyltransferase (ACAT)
- Utilization of cholesterol for synthesis of other steroids, such as hormones, or bile acids in the liver

National Cholesterol Education Program (NCEP)

mg/dL	Total cholesterol	HDL cholesterol	LDL cholesterol	Triglyceride
Desirable	< 200	>40	<130	<150
Borderline	200-239		130-159	150-199
High risk	>240		>160	>200

Dyslipoproteinemia

- Abnormally high or low lipoprotein concentrations
- Abnormalities in the composition of the lipoprotein particles
- Include hypolipoproteinemia and hyperlipoproteinemia

Primary hyperlipoproteinemia :

- A group of genetic disorders of the lipid transport proteins in the blood
- Abnormally elevated levels of cholesterol, triglycerides, or both in the serum of affected patients

Example:

Hypertriglyceridemia: Lipoprotein lipase deficiency, apo CII deficiency

Hypercholesterolemia : LDL receptor deficiency, apo B100 deficiency

Secondary hyperlipoproteinemia:

- Variety of exogenous factors or clinical disorders i.e. obesity, type2 diabetes mellitus, hypothyroidism

Familial hypercholesterolemia

- Genetic disease caused by a defective of LDL-receptor gene
- One of the well-understood risk factors of atherosclerosis
- Clinical: premature coronary heart disease, xanthomas (visible subcutaneous lipid deposits, typically yellow or brownish)
- Metabolic: increase level of LDL

This image was from Varghese MJ. Familial hypercholesterolemia: A review. Ann Pediatr Cardiol 2014;7(2):107–117.





➢ Heterozygous

- Have one abnormal copy the LDL receptor gene
- Premature cardiovascular disease at the age of 30 to 40
- Normally treated with some drug (statins) or other hypolipidemic agents that lower cholesterol levels

➢Homozygous

- Have two abnormal copies of LDL receptor gene
- Severe cardiovascular disease in childhood